CS 306: Wireless Networks

Dr. Ayan Mondal

Assistant Professor

Dept. of CSE, IIT Indore

https://www.iiti.ac.in/people/~ayanm/

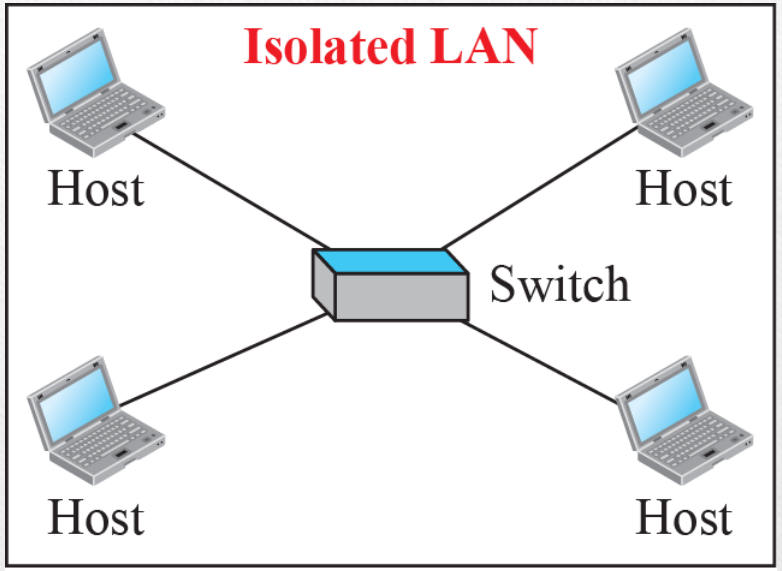
Outline

- IEEE 802.11

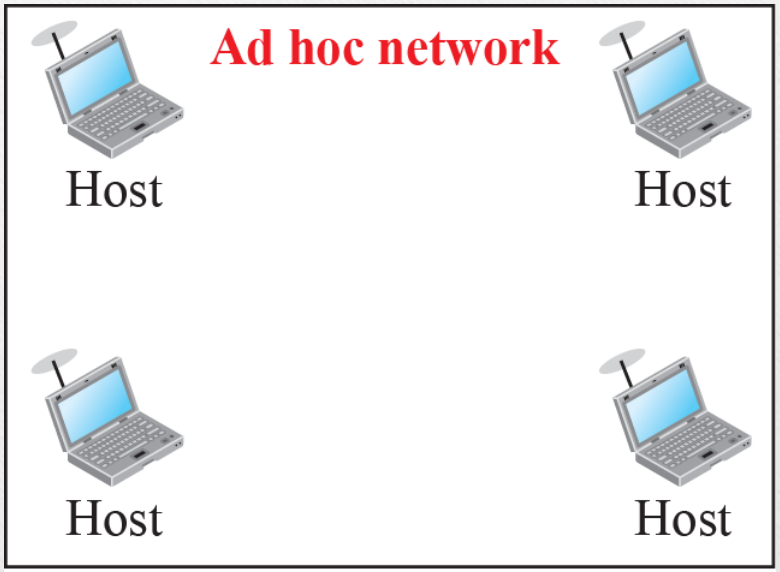
Introduction

- Wireless communication is one of the fastest-growing technologies.
- The demand for connecting devices without the use of cables is increasing everywhere.
- Wireless LANs can be found on college campuses, in office buildings, and in many public areas.

Architectural Comparison

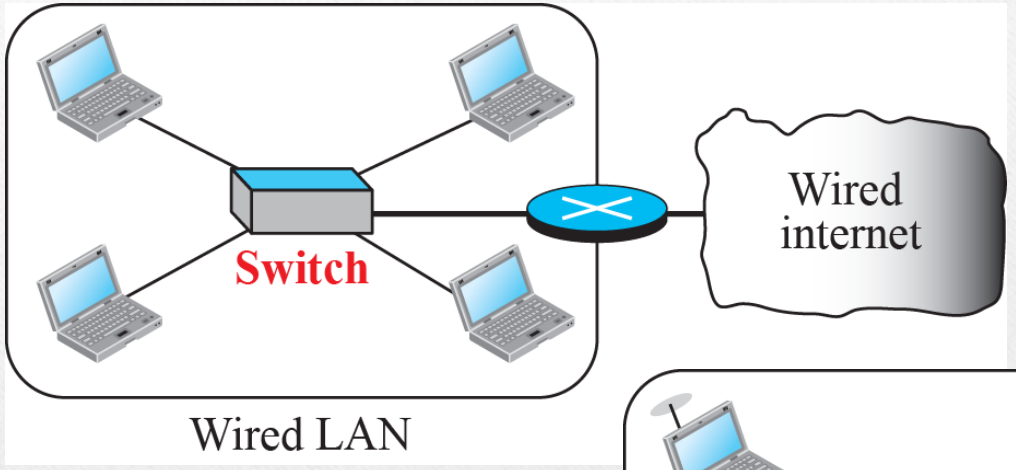


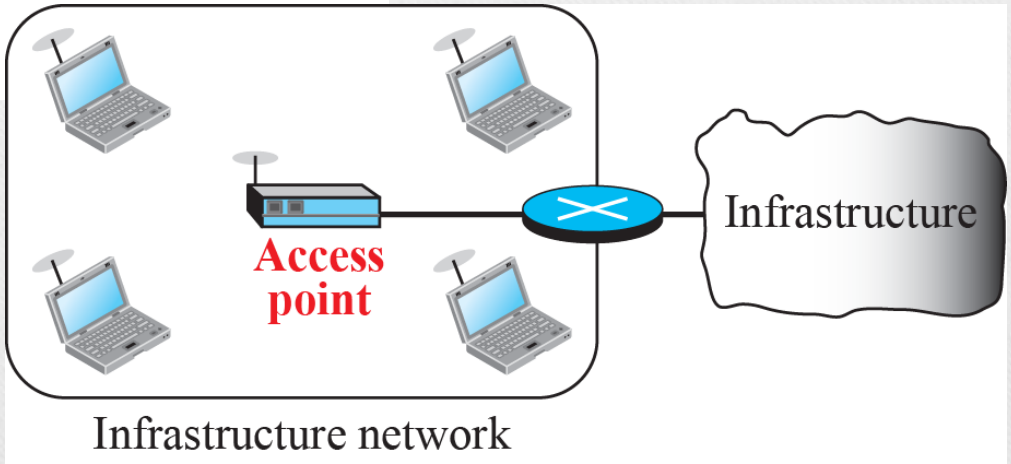
Wired



Wireless

Connection of a wired LAN and a wireless LAN

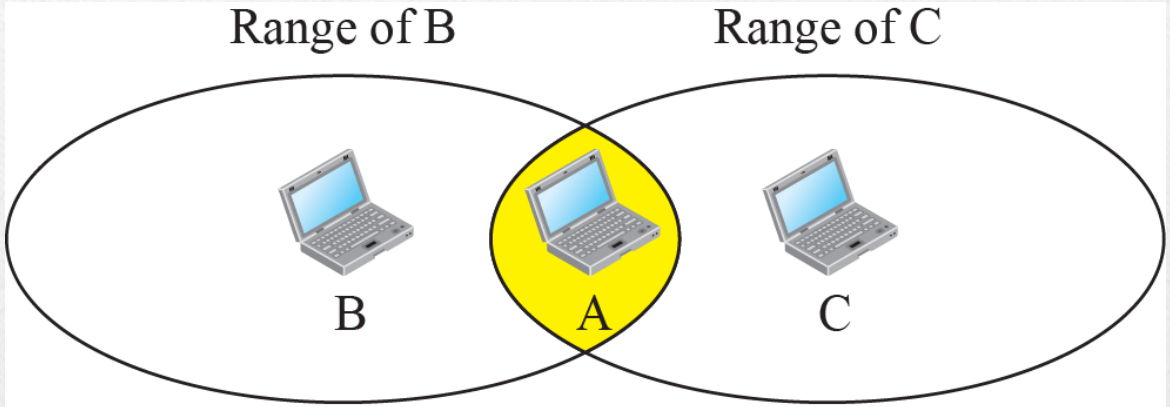




Access Control

- The CSMA/CD algorithm does not work in wireless LANs for three reasons:
  - Wireless hosts do not have enough power to send and receive at the same time.
  - The hidden station problem prevents collision detection
  - The distance between stations can be great.

Hidden station problem



a. Stations B and C are not in each other's range.

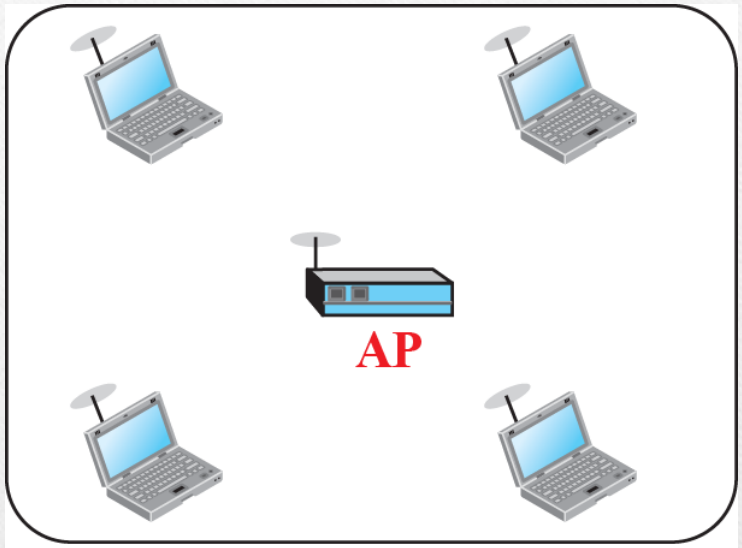
IEEE 802.11 PROJECT

- IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data-link layers. It is sometimes called wireless Ethernet.

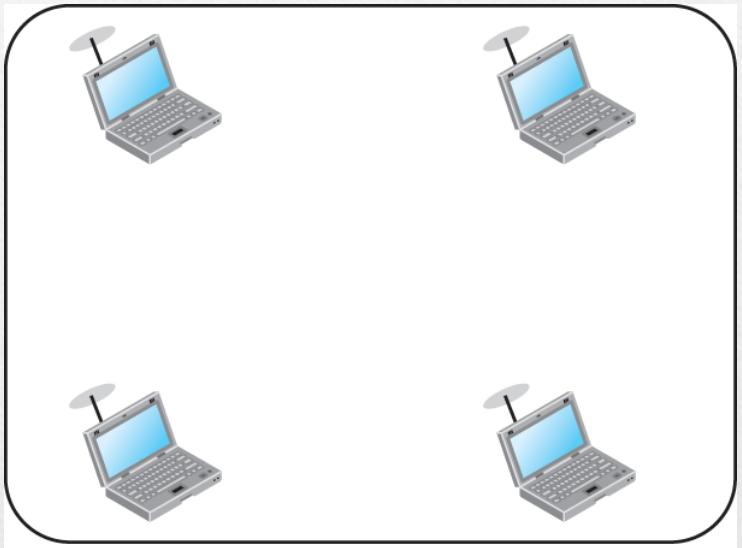
Architecture

- The standard defines two kinds of services:
  - Basic service set (BSS)
  - Extended service set (ESS)

Basic service sets (BSSs)

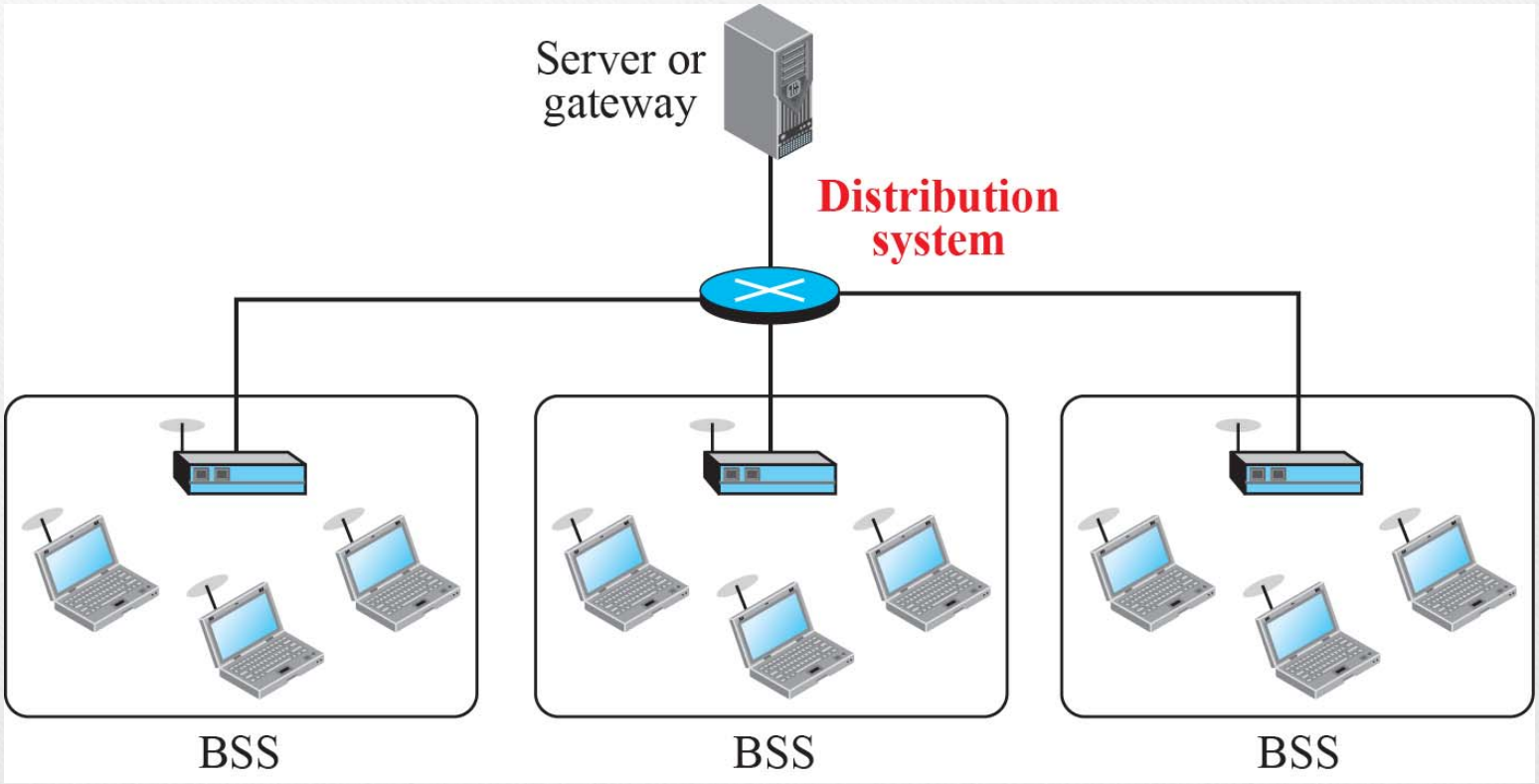


Infrastructure BSS



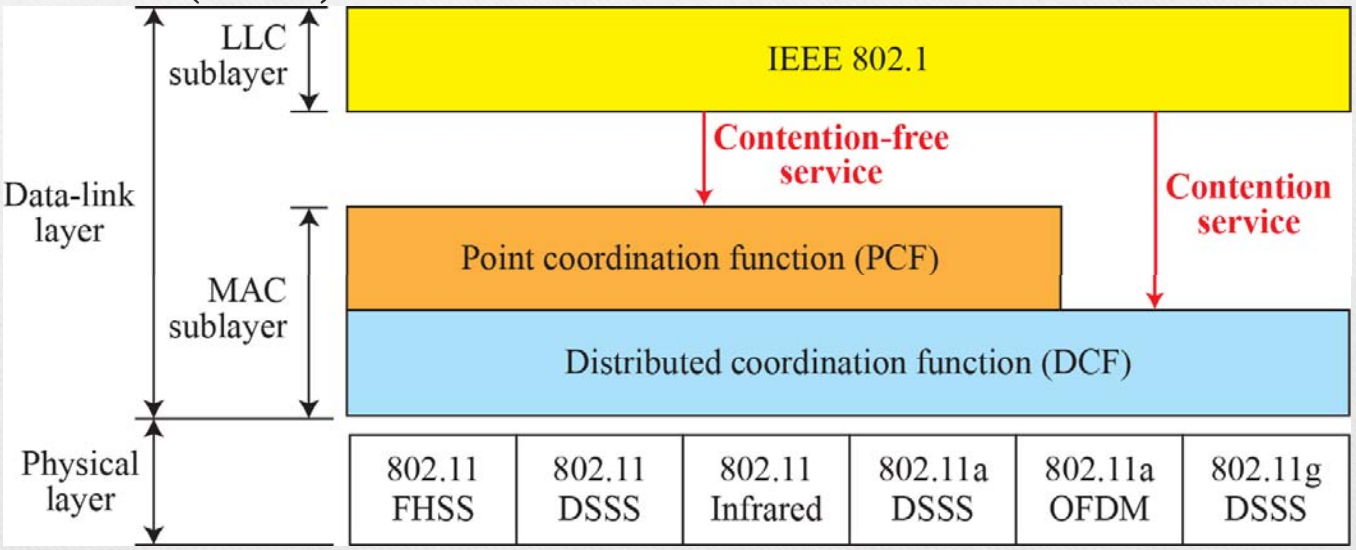
Ad hoc BSS

Extended Service Set (ESS)



MAC Sublayer

- IEEE 802.11 defines two MAC sublayers: the distributed coordination function (DCF) and point coordination function (PCF).



Frame format

2 bytes	2 bytes	6 bytes	6 bytes	6 bytes	2 bytes	6 bytes	0 to 2312 bytes	4 bytes
FC	D	Address 1	Address 2	Address 3	SC	Address 4	Frame body	**FCS**

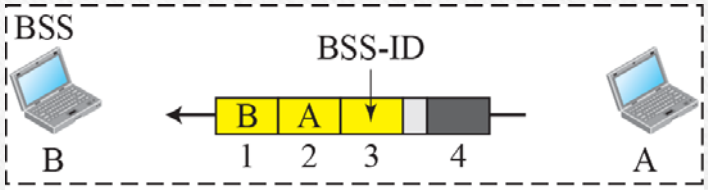
Protocol version	Type	Subtype	To DS	From DS	More frag	Retry	Pwr mgt	More data	WEP	Rsvd
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Addressing Mechanism

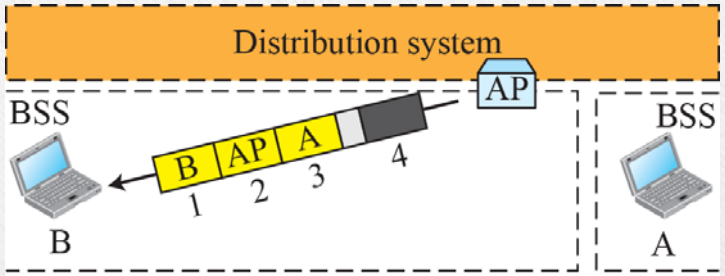
- The IEEE 802.11 addressing mechanism specifies four cases, defined by the value of the two flags in the FC field, To DS and From DS.
- Each flag can be either 0 or 1, resulting in four different situations.

To DS	*From DS*	*Address 1*	*Address 2*	*Address 3*	*Address 4*
0	0	Destination	Source	BSS ID	N/A
0	1	Destination	Sending AP	Source	N/A
1	0	Receiving AP	Source	Destination	N/A
1	1	Receiving AP	Sending AP	Destination	Source

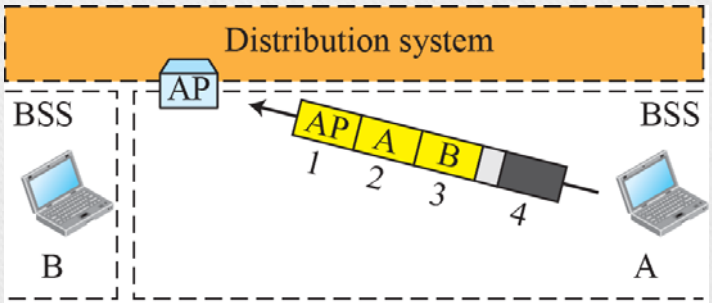
Addressing mechanisms



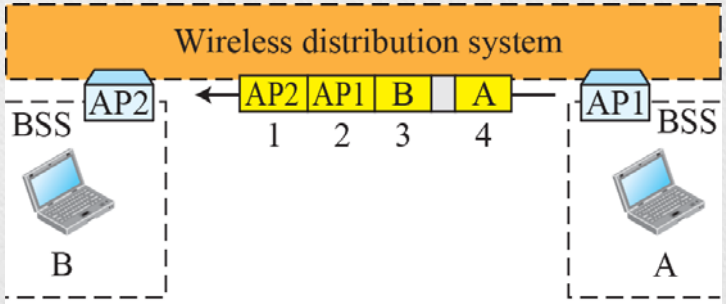
a. Case 1



b. Case 2



c. Case 3



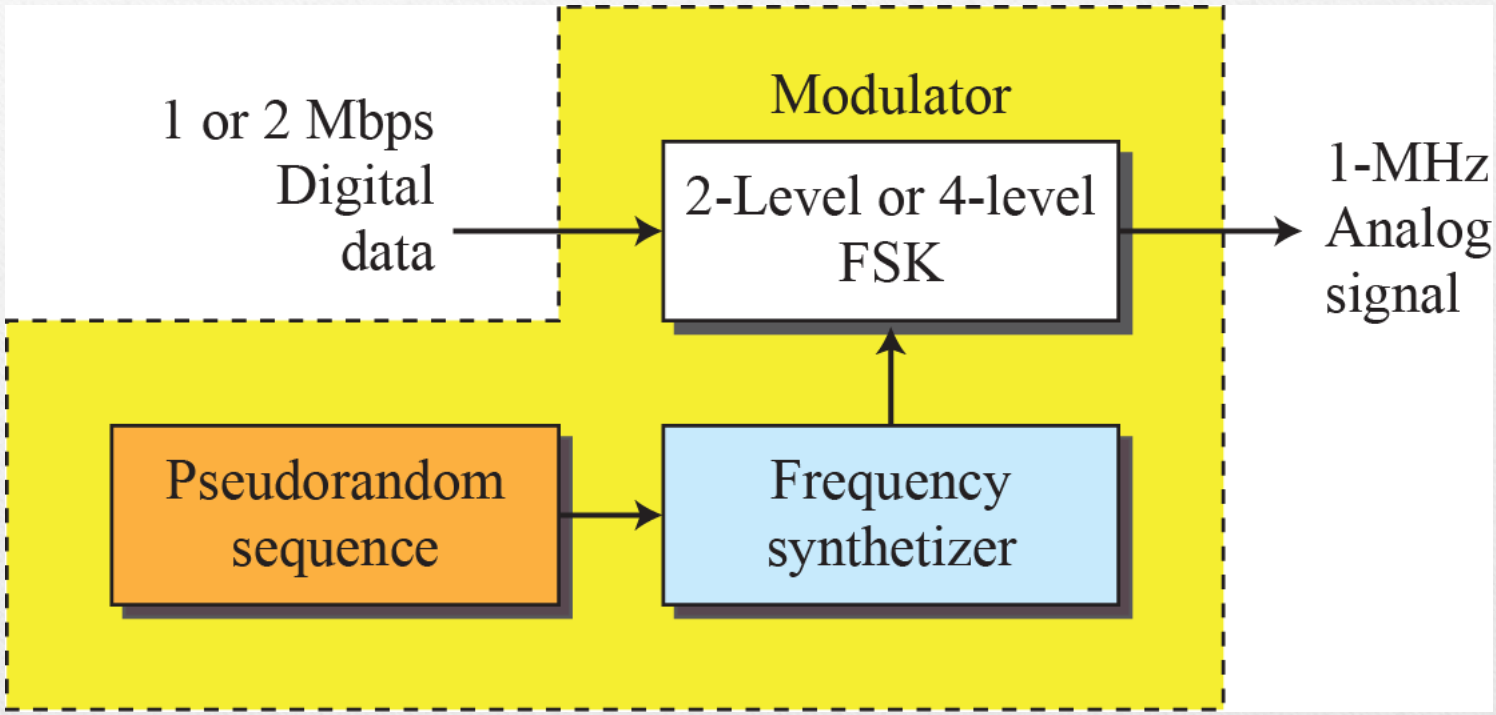
d. Case 4

Physical Layer

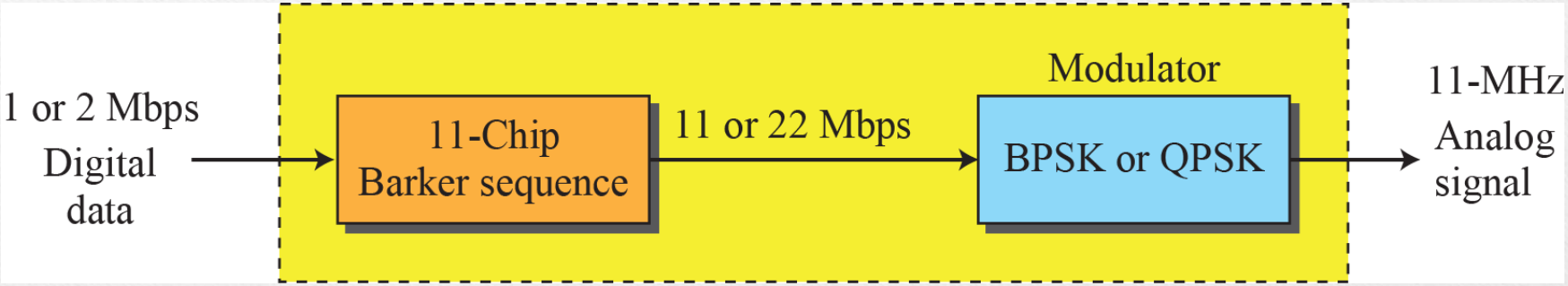
- All implementations, except the infrared, operate in the industrial, scientific, and medical (ISM) band, which defines three unlicensed bands in the three ranges 902–928 MHz, 2.400–4.835 GHz, and 5.725–

IEEE	*Technique*	*Band*	*Modulation*	*Rate (Mbps)*
802.11	FHSS	2.400–4.835 GHz	FSK	1 and 2
	DSSS	2.400–4.835 GHz	PSK	1 and 2
	None	Infrared	PPM	1 and 2
802.11a	OFDM	5.725–5.850 GHz	PSK or QAM	6 to 54
802.11b	DSSS	2.400–4.835 GHz	PSK	5.5 and 11
802.11g	OFDM	2.400–4.835 GHz	Different	22 and 54
802.11n	OFDM	5.725–5.850 GHz	Different	600

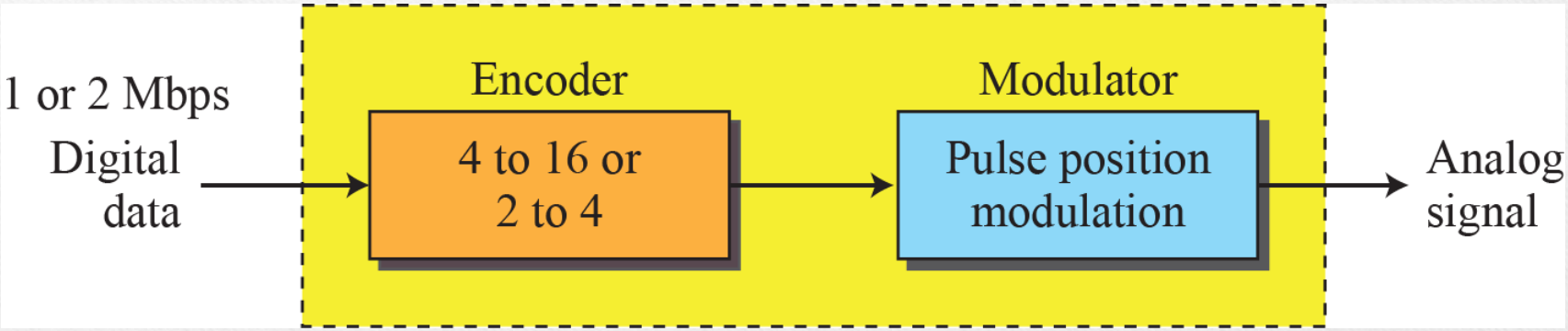
Physical layer of IEEE 802.11 FHSS



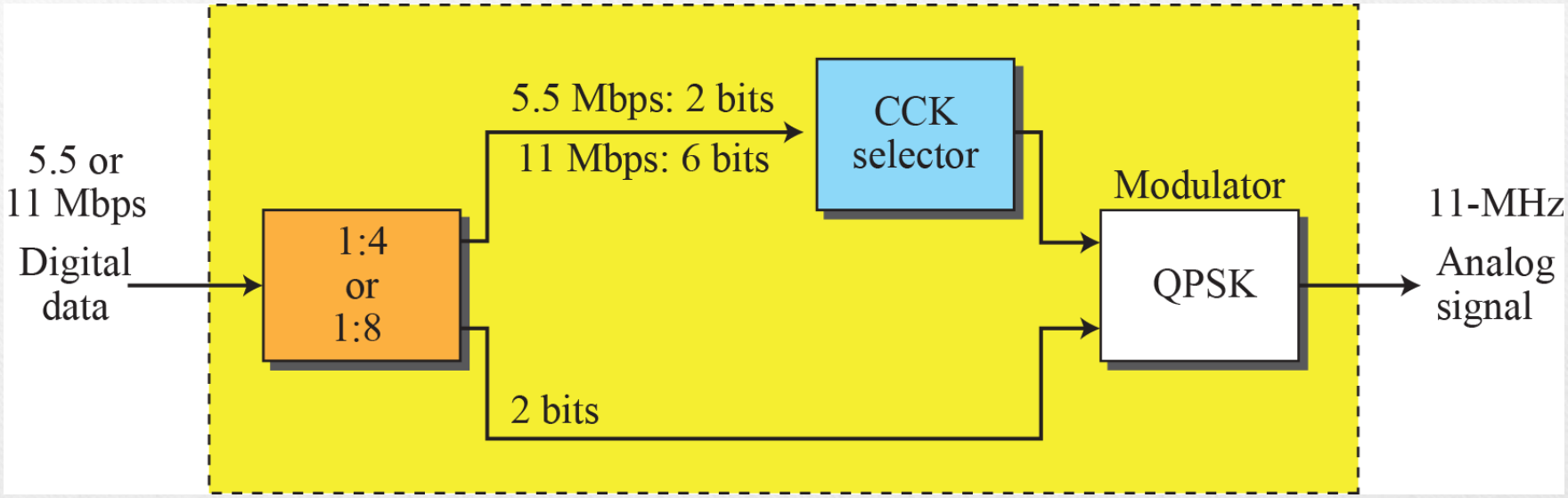
Physical layer of IEEE 802.11 DSSS



Physical layer of IEEE 802.11 infrared



Physical layer of IEEE 802.11b



Next Class

- Bluetooth

?

ayanm@iiti.ac.in

Room No. 407, POD-1A